

Grocery Retail Demand | Overview

What is driving demand across selected stores and product families?



89.21M

Total Demand

150.69K

Average Daily Demand

9.35M

Total Transactions

8

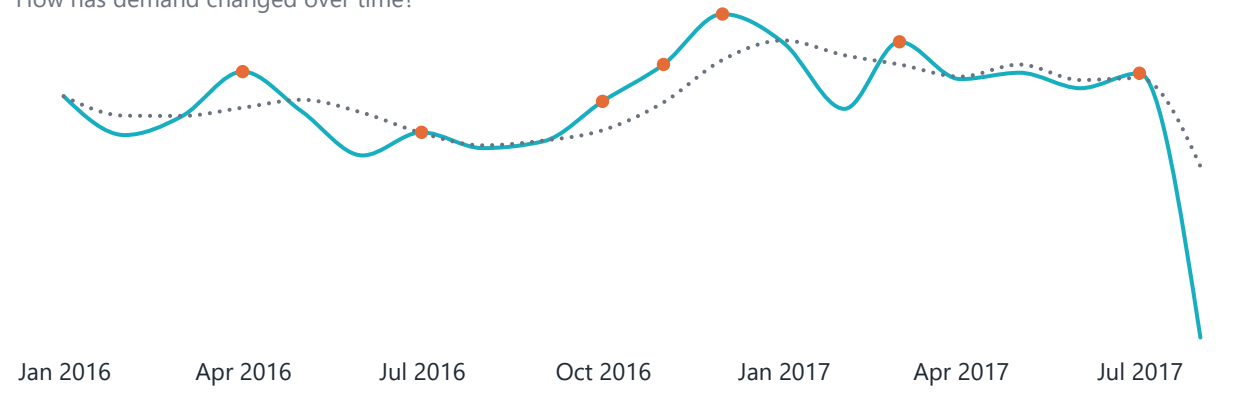
Number of Product Families

75.85%

Top 3 Family Share

Monthly Demand Trend

How has demand changed over time?



Demand Matrix by Store and Product Family

Where is demand concentrated across stores and categories?

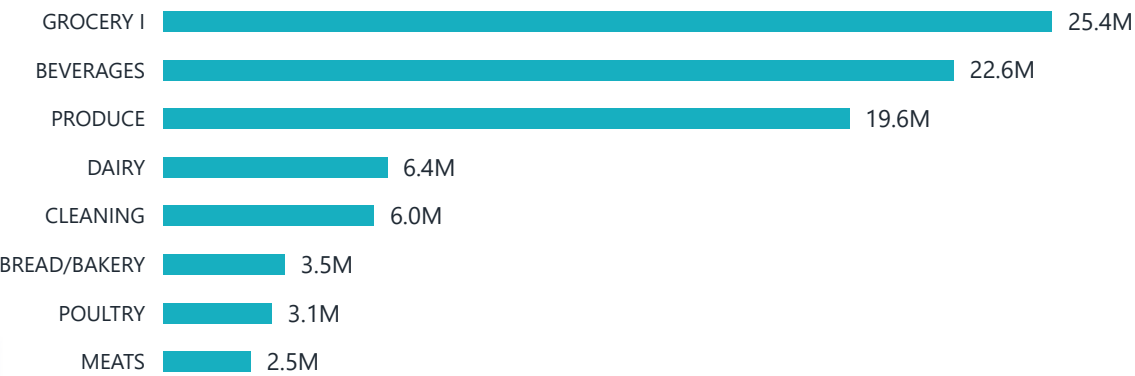
Family	25	44	45	47	49
GROCERY I	1,043,526.0	6,362,273.0	6,740,239.4	6,317,761.0	4,960,570.0
BEVERAGES	1,287,268.0	6,129,689.0	5,565,028.0	5,162,845.0	4,470,933.0
PRODUCE	883,251.7	6,055,693.4	3,998,917.2	3,717,193.8	4,968,573.3
DAIRY	322,872.0	1,684,682.0	1,477,545.0	1,429,745.0	1,493,053.0
CLEANING	265,708.0	1,528,944.0	1,595,889.0	1,454,493.0	1,188,017.0
BREAD/BAKERY	189,786.5	992,843.2	787,234.7	734,913.3	777,956.9
POULTRY	93,618.7	791,421.6	826,617.2	748,305.2	651,381.0
MEATS	112,681.5	573,922.1	735,639.7	570,606.0	518,748.1
Total	4,198,712.4	24,119,468.2	21,727,110.2	20,135,862.3	19,029,232.5

KEY INSIGHT

The top three product families — GROCERY I, BEVERAGES, and PRODUCE — contribute 75.85% of total demand. Store 44 generated the highest demand among the selected stores. These patterns suggest that replenishment planning should prioritize high-volume categories and high-demand stores.

Demand by Product Family

Which product families drive the most demand?



Demand by Store

Which stores generate the highest demand?



Demand Forecasting Performance | Forecast

How accurately can we forecast short-term demand for priority grocery categories?



30

Forecast Horizon (Days)

3

Families Forecasted

7-Day Seasonal Naive

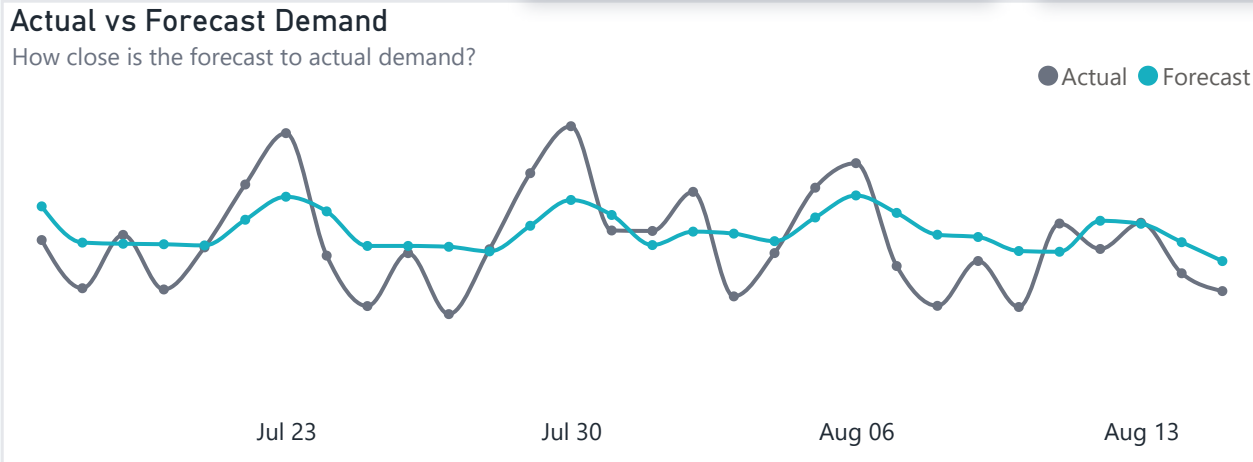
Best Model

7.65

Best MAPE (%)

3.02M

Total 30-Dav Forecast



Forecast Summary by Product Family

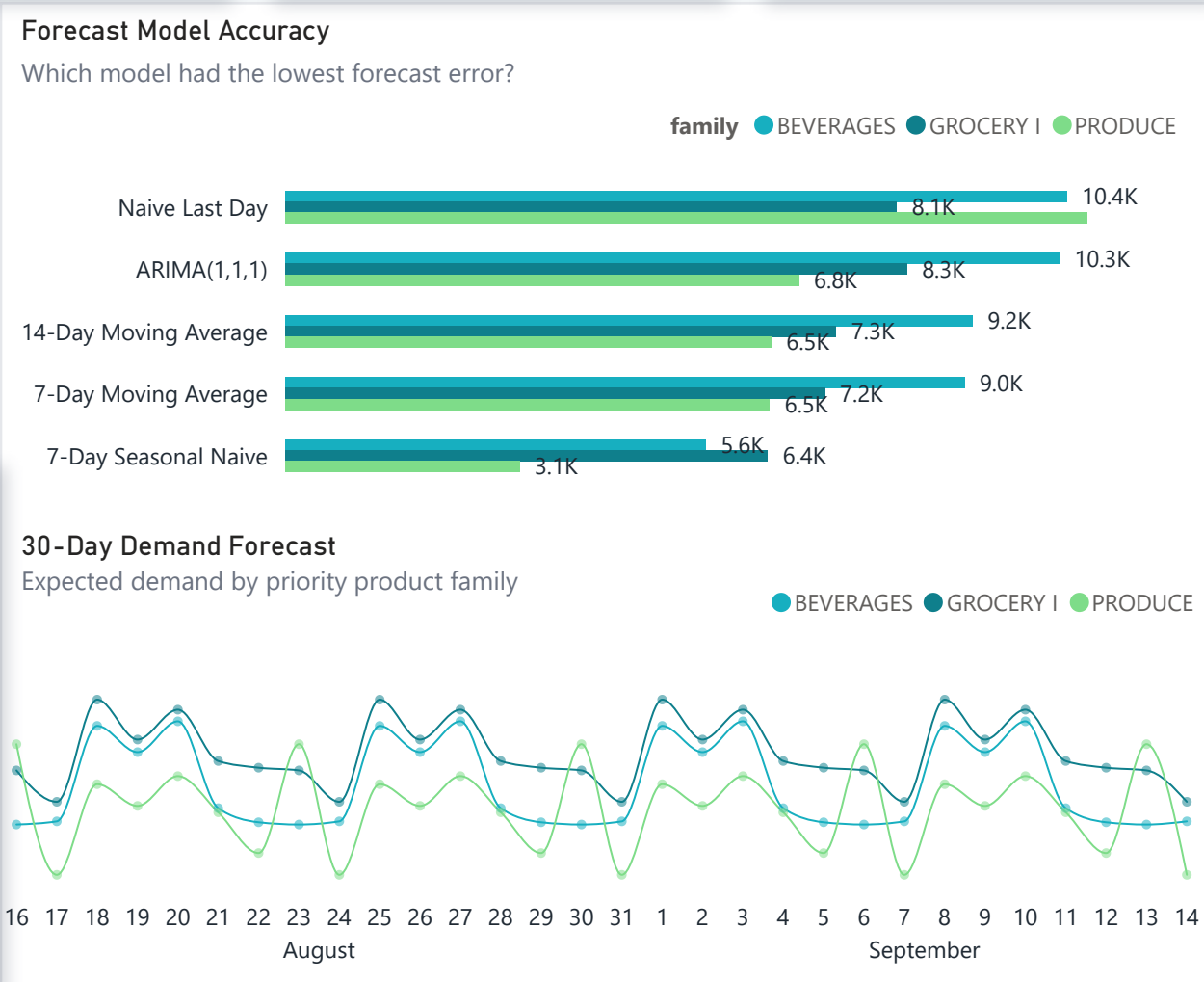
Family	Total 30-Day Forecast	Avg Daily Forecast	Min Daily Forecast	Max Daily Forecast
GROCERY I	1113124.0	37104.1	31135	43477
BEVERAGES	995093.0	33169.8	28388	40858
PRODUCE	913832.1	30461.1	22314	38096

Model Performance Details

Family	Model	MAE	RMSE	MAPE
BEVERAGES	14-Day Moving Average	7589.6	9158.3	21.2
BEVERAGES	7-Day Moving Average	7519.9	9049.0	20.8
BEVERAGES	7-Day Seasonal Naive	4504.8	5600.4	12.6
BEVERAGES	ARIMA(1.1.1)	9115.7	10313.2	26.4

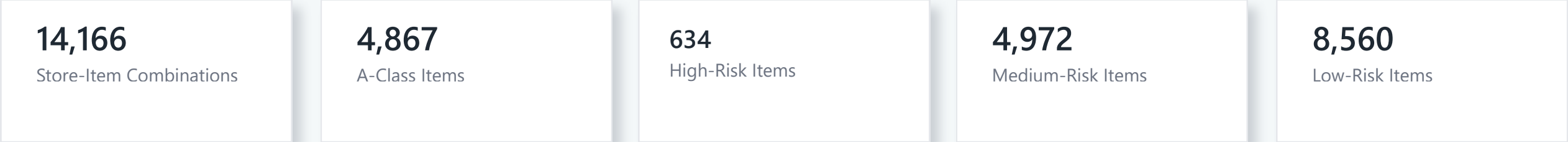
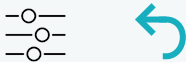
KEY INSIGHT

The 7-Day Seasonal Naive model performed best for all three high-priority product families. ARIMA(1,1,1) was tested as an advanced model, but it did not outperform the weekly seasonal baseline. This suggests that weekly demand patterns are important for short-term grocery demand forecasting.



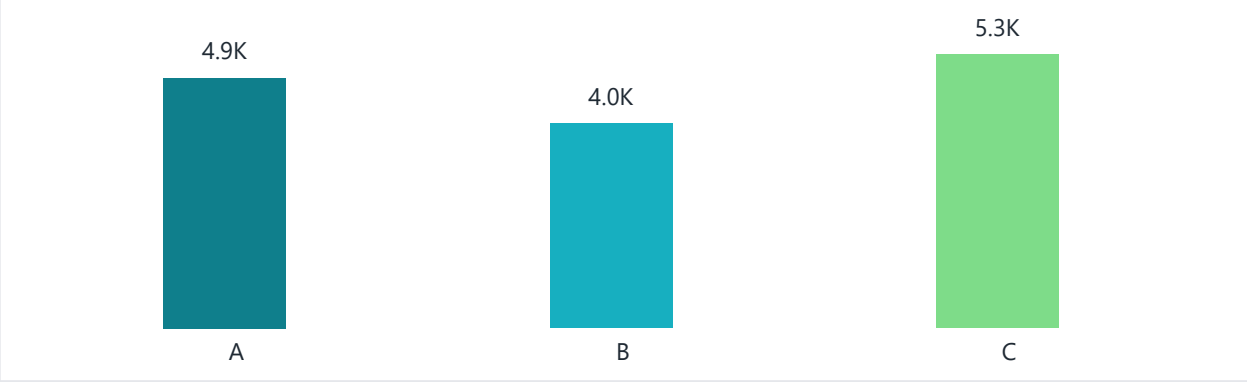
Inventory Optimization & Replenishment Risk

Which products need closer replenishment monitoring?



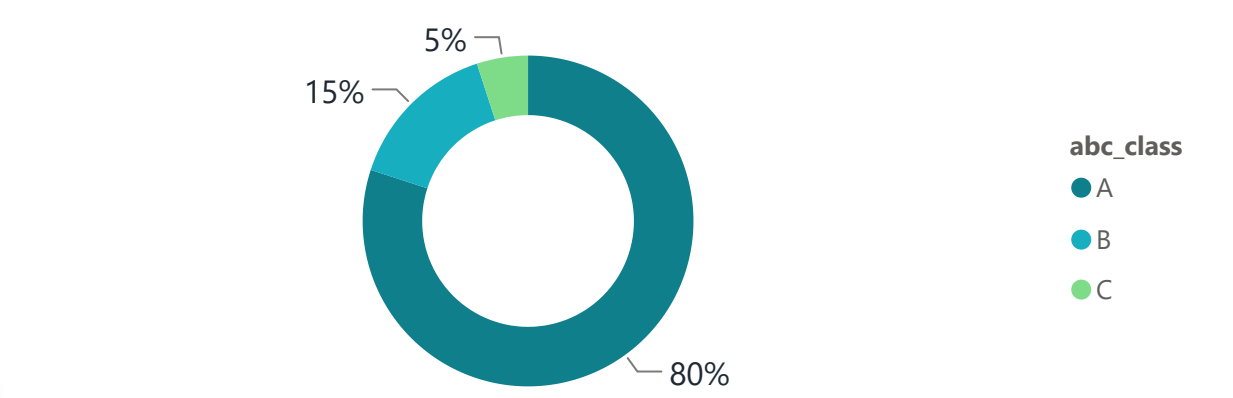
ABC Classification Distribution

How many store-item combinations fall into each priority class?



Demand Share by ABC Class

How much demand is represented by each inventory class?



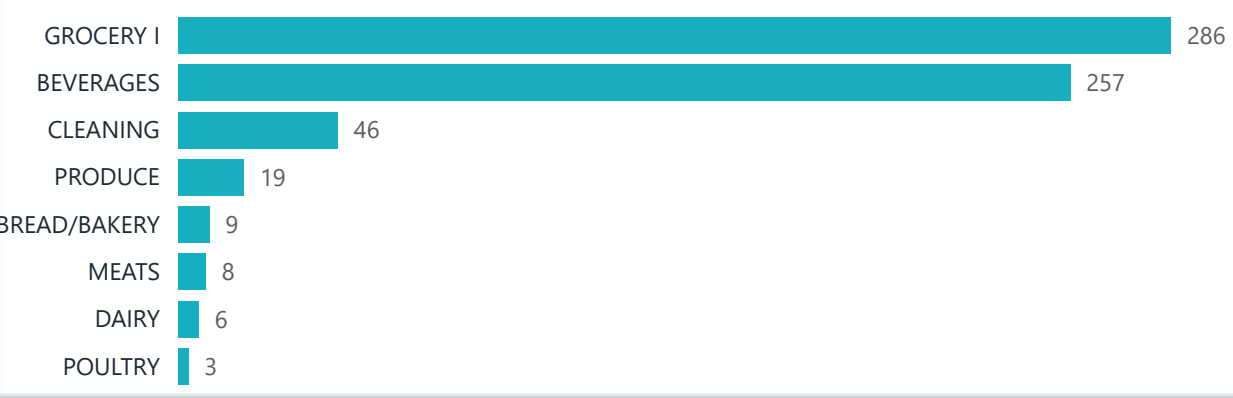
Safety Stock vs Reorder Point

Which high-risk items require higher buffer stock and reorder levels?



High-Risk Items by Product Family

Which product families contain the most high-risk items?



KEY INSIGHT

A-items account for approximately 80% of total demand, while 634 store-item combinations were classified as high replenishment risk. GROCERY I and BEVERAGES contain the highest number of high-risk items, suggesting that replenishment monitoring should prioritize these high-volume categories.